

AN EIGHT-SEED PROOF OF THE SHORTEST THREE-OCTAD LEECH TRACE INVERSE

THE CLANKERS: RESEARCH DRAFT

ABSTRACT. Three disjoint Golay octads define integral traces $P : \Lambda \rightarrow \mathbb{Z}^3$. We prove that every trace lifts and compute the shortest squared norm as $\|t\|^2/4 + \mu(t \bmod 4)$, with $0 \leq \mu \leq 15/4$. Eight norm-four seeds, a norm-six dodecad, explicit phase permutations and block signs supply all reduced fibres. Translation proves the result for every integer input. The proof retains the kernel coordinate and does not infer ES existence from linear surjectivity.

1. THE FIXED COORDINATE MODEL

Write $\Omega = \mathbb{F}_{23} \cup \{\infty\}$, serializing ∞ as coordinate 23. If Q is the set of nonzero squares modulo 23, let $\mathcal{C} \subset \mathbb{F}_2^{24}$ be the span of the 23 supports $(a + Q) \cup \{\infty\}$. A word is represented by the binary integer whose i th bit is its i th coordinate. The twelve words

85335e, 5606be, 2ca9ae, 11fe26, 0f55e2, 046f3e,
02379f, 0149a6, 00a4d3, 0042de, 003e4a, 001f25

form a basis; the notation is hexadecimal. Row reduction of the defining supports gives this basis. Enumerating its 2^{12} sums gives weights 0, 8, 12, 16, 24 with multiplicities 1, 759, 2576, 759, 1. This is the projective Golay code [Chapman(1997)]. We use the Leech model [Brouwer(n.d.), Theorem 1.4]

$$\Lambda = \left\{ x/\sqrt{8} : \begin{array}{l} x \in \mathbb{Z}^{24}, \quad x_i \equiv m \pmod{2} \text{ for one } m \in \{0, 1\}, \\ \sum_i x_i \equiv 4m \pmod{8}, \quad ((x_i - m)/2 \bmod 2)_i \in \mathcal{C} \end{array} \right\}. \quad (1)$$

Thus Λ is even, unimodular and has minimum squared norm four. These properties and the lattice itself are inherited, not claimed as new.

The three disjoint octads are

$$\begin{aligned} B_0 &= \{0, 1, 3, 4, 7, 8, 16, 23\}, \\ B_1 &= \{10, 12, 15, 18, 19, 20, 21, 22\}, \\ B_2 &= \{2, 5, 6, 9, 11, 13, 14, 17\}. \end{aligned}$$

Put $w_i = 2\mathbf{1}_{B_i}/\sqrt{8}$ and $P_i(x/\sqrt{8}) = \sum_{j \in B_i} x_j/4$. The matrices

$$C = \begin{pmatrix} 12 & 5 \\ 1 & 12 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 7 \\ 3 & 22 \end{pmatrix}, \quad T = \begin{pmatrix} 4 & 11 \\ 16 & 4 \end{pmatrix}$$

act on Ω by fractional linear transformations over $\mathbb{F}_{23}$ and permute lattice coordinates. They preserve $\mathcal{C}$; this can be checked on its twelve basis words. In projective action $T^4 = C$, and C cycles B_0, B_1, B_2 . All vector actions below are these coordinate permutations, not octonion-coordinate permutations.

2. THE SHORTEST INVERSE

For $r \in \{0, 1, 2, 3\}^3$, set $a_i(r) = r_i$ for $r_i \leq 2$ and $a_i(r) = -1$ for $r_i = 3$. Put $d(r) = \sum_i a_i(r)^2$.

Theorem 1. *Every $t \in \mathbb{Z}^3$ is $P\lambda$ for some $\lambda \in \Lambda$, and*

$$\boxed{\min_{P\lambda=t} \|\lambda\|^2 = \frac{\|t\|^2}{4} + \mu(t \bmod 4)}, \quad \mu(r) = \begin{cases} 0 & r = 000, \\ 3 & r = 222, \\ 4 - d(r)/4 & \text{otherwise.} \end{cases}$$

In particular $0 \leq \mu(r) \leq 15/4$. The inverse below attains this minimum at every integer input, including negative and zero traces.

Proof. We give eight norm-four seeds, rather than require a reader to trust a census of all minimal vectors. For subsets $N \subset S \subset \Omega$ define the numerator $E(S, N) = 2\mathbf{1}_S - 4\mathbf{1}_N$. Define $O(j, \epsilon, N)$ to have coordinate 3ϵ at j and $1 - 2\mathbf{1}_N(i)$ elsewhere. Entries in the following table are numerators, to be divided by $\sqrt{8}$:

Trace	Seed
100	$O(0, -1, \{2, 5, 6, 10, 12, 13, 15, 22\})$
110	$E(\{0, 2, 7, 11, 12, 16, 19, 23\}, \{0, 2\})$
111	$O(23, 1, \{0, 1, 2, 3, 9, 12, 21\})$
200	$E(\{0, 1, 3, 4, 7, 8, 16, 23\}, \{0, 1\})$
210	$O(16, 1, \{0, 2, 5, 6, 11, 18, 22\})$
211	$E(\{7, 8, 12, 13, 14, 16, 18, 23\}, \emptyset)$
220	$E(\{4, 7, 16, 18, 19, 20, 21, 23\}, \emptyset)$
221	$O(2, -1, \emptyset)$

Each seed satisfies (1), has exactly the indicated trace and squared norm four; these are eight explicit integer/code calculations. The matrix

$$R = \begin{pmatrix} 5 & 6 \\ 19 & 0 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{F}_{23})$$

preserves the code and exchanges B_1, B_2 while fixing B_0 . Together with C it realizes all permutations of the three traces. Membership of $RC = C$ is verified on the twelve code basis words.

Independently changing the signs on each entire block also preserves (1). Indeed, write $x = m\mathbf{1} + 2c + 4z$. Self-orthogonality gives $|c \cap B_j|$ even; hence $\sum_{i \in B_j} x_i$ is divisible by four, and changing those signs preserves the numerator-sum congruence. The binary word changes from c to $c + m\mathbf{1}_{B_j}$, still a codeword. The squared norm is unchanged. Thus permutations and block signs transport the eight seeds to all 62 nonzero reduced traces except 222.

Every nonzero vector of Λ has squared norm at least four, proving optimality at those 62 reduced traces. At 000 use zero. At 222, the numerator sum is 24, so the common parity in (1) is even. Write $x = 2y$. Each block sum of y is four, and

$$\|x/\sqrt{8}\|^2 = \frac{1}{2} \sum_i y_i^2 \geq \frac{1}{2} \sum_i |y_i| \geq 6.$$

Equality is attained by $E(S, \emptyset)/\sqrt{8}$ for

$$S = \{0, 3, 4, 5, 6, 8, 9, 10, 12, 13, 15, 18\},$$

a Golay dodecad with four entries in each block. This proves the reduced minimum six.

Finally, $(w_i, w_j) = 4\delta_{ij}$. The component of any vector in the real span of the w_i is

$$q(t) = \frac{1}{4} \sum_i t_i w_i, \quad \|q(t)\|^2 = \|t\|^2/4.$$

Translation by w_i changes only the i th trace by four and leaves the perpendicular component unchanged. For a reduced representative ν_r just constructed, the vector

$$\lambda_*(t) = \nu_r + \sum_i \frac{t_i - a_i(r)}{4} w_i, \quad r = t \bmod 4$$

is integral, has trace t , and has the same minimal perpendicular norm as ν_r . This proves the all-input formula and both the asserted minimum and its attainment. $\square$

3. FIBRES, RETAINED STATE AND ARITHMETIC SCOPE

The three norm-four vectors b_0, Cb_0, C^2b_0 , with

$$b_0 = \frac{1}{\sqrt{8}}(-3, 1, -1, 1, 1, -1, -1, 1, 1, 1, 1, 1, -1, 1, 1, 1, 1, -1, -1, -1, -1, 1, 1),$$

have respective traces $(1, 0, 0), (0, 1, 0), (0, 0, 1)$. Put $\sigma(t) = \sum_i t_i C^i b_0$ and $K = \ker P$. Then

$$(t, k) \mapsto \sigma(t) + k, \quad \lambda \mapsto (P\lambda, \lambda - \sigma(P\lambda))$$

are exact inverse group homomorphisms between $\mathbb{Z}^3 \oplus K$ and Λ . This is not an orthogonal direct product. The rank of K is 21 and its determinant is 64: $L = \bigoplus_i \mathbb{Z}w_i$ is primitive by pairing with the $C^i b_0$, and $\Lambda/(L \oplus K) \cong (\mathbb{Z}/4)^3$, so unimodularity gives $\det L \det K = 64^2$ with $\det L = 64$.

For a symmetry g descending to ρ_g on the traces, the full inverse action is

$$(t, k) \mapsto (\rho_g t, gk + g\sigma(t) - \sigma(\rho_g t)).$$

The kernel correction cannot be omitted merely because a shortest representative was selected. If a symmetry does not descend to the traces, the full coordinate vector or a larger stable readout is required.

For $F_p(t) = 4t_0t_1t_2 - p \sum_{i < j} t_it_j$, surjectivity identifies positive integral ES solutions with lattice vectors satisfying $P\lambda > 0$ and $F_p(P\lambda) = 0$. It does not prove that this additional cubic locus is occupied. One immediate exact restriction is $r \neq 222$ for odd p : that residue gives $F_p(t) \equiv 4 \pmod{8}$.

4. REPRODUCIBILITY AND ATTRIBUTION

`verify_shortest.py`, using only Python's standard library and the accompanying coordinate routines, verifies the eight seeds, the dodecad, both phase generators, and all 64 transported representatives. The all-integer conclusion uses translation and the Leech minimum, not enumeration up to an unspecified height. `reconstruct_checked.py` also verifies the returned vector's exact height against the formula before describing it as shortest.

The Leech model, Golay code and their symmetries are inherited [Brouwer(n.d.), Chapman(1997)]; Wilson's three-octonion construction is related background [Wilson(2009)]. This note isolates the explicit three-trace height formula and provides a shorter seed proof than the preceding full-shell computation. A scoped content search did not identify an exact earlier statement of this formula, but no historical-priority or new-lattice claim is made. No formal theorem-prover build or independent review is claimed.

REFERENCES

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- [Wilson(2009)] Wilson, R. A. (2009). *Octonions and the Leech lattice*, revision of 20 March, Sections 2–4. https://maths.qmul.ac.uk/~raw/pubs_files/octoLeech1rev.pdf.